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Rapid and nondestructive analysis of the ethylene content of propylene/ethylene copolymer by near-infrared spectroscopy

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Abstract

Near Infrared (NIR) spectroscopy has been successfully implemented for the rapid and nondestructive analysis of ethylene content in flake and pellet of propylene/ethylene copolymer. The change of copolymer structure due to the variation of ethylene content during copolymerization reaction resulted in systematic NIR spectral variations. By using a partial least squares (PLS) regression combined with second derivative algorithm, relevant spectral information was successfully extracted and correlated with conventional infrared (IR) method. The error of NIR spectroscopic method for determination of ethylene content was comparable to the reference IR method. With the help of rapid and nondestructive natures of NIR spectroscopy, ethylene content can be measured less than 2 min without a physical sample pretreatment and the copolymerization process can be optimally controlled by continuously reflecting the process status from the NIR measurement. © 1998 Elsevier Science B.V. All rights reserved.

Keywords: Near infrared spectroscopy; NIR; Propylene/ethylene copolymer; Copolymerization; Ethylene content; Pellet; Flake

1. Introduction

Propylene/ethylene copolymer is one of the most popular polymer serving as a basic feed material for injection molding industries [1]. Generally it is produced from the copolymerization of propylene and ethylene with necessary catalysts. There are several types of polymerization process from different licensers depend on phases of reaction and types of catalyst [2]. One of the most important analytical parameter for the quality of final copolymer is the ethylene content. An increase of ethylene relative to propylene during copolymerization results in more CH_2 groups relative to CH_3 groups in final copolymer structure. The schematic structural changes based on ethylene content are presented in Fig. 1. The structural change causes the variation of physical properties that increasing the ethylene content in the copolymer causes an increment in the impact resistance and shift of the melting curve to lower temperature with line broadening [3]. Since the properties of propylene/ethylene copolymer are greatly influenced by the ethylene content, the rapid and accurate determination is essential to control the process and keep the consistency in final product quality.

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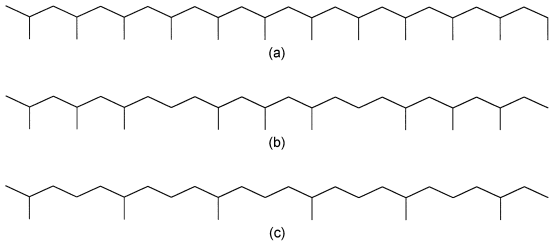

Fig. 1. The schematic structural changes depend on (a) zero, (b) medium, and (c) high ethylene content in propylene/ethylene copolymer.

Conventionally infrared (IR) transmission spectroscopy has been extensively used in routine compositional analysis of polymer [4,5]. The determination of ethylene content in blended or copolymerized polymers using IR spectroscopy is mostly based on a ratio of peak height or area in the same IR spectrum. Several methods using different IR peaks have been discussed for the determination of composition in blended copolymer (polyethylene and polypropylene) [6,7]. Even though IR spectroscopy has been widely utilized, polymer samples should be pretreated to make a thin plastic sheet suitable to IR transmission by heating over 150°C, hot pressing, and cooling. This pretreatment step results in longer analysis and labor time, and occasionally alters the physical properties of sample which could degrade the accuracy of measurement [4,8]. Additionally this type of sampling is restricted occasionally because some polymers are infusible and insoluble without degradation.

Near-infrared (NIR) spectroscopy [9,10] is an excellent alternative analytical method for the analysis of polymer without a sample pretreatment step and degradation. The wide applicability and potential of NIR spectroscopy for various polymer analyses have been recognized [11]. NIR spectrum also contains information related polymer properties [12] such as composition, conformation, and crystallinity, therefore it can be widely applied for various polymer analyses in conjunction with chemometric calibration methods. The integration of optical fiber and NIR spectrometer allows remote and on-line process analyses easily such as monitoring of isocyanate in the urethane polymerization [13], vinyl acetate in poly(ethylene vinyl acetate) polymerization [14], and compositions of polymer blends (polypropylene/ethylene vinyl acetate) [15] during extrusion. The polymer composition was determined by a peak ratio using a preprepared polymer sheet with NIR transmission spectroscopy [7,16]; however, it required the same pretreatment step as IR method. Additionally the reconstruction of IR spectra from NIR spectra by a using partial least squares (PLS) has been attempted, and then reconstructed IR spectra were used to determine the composition in blended copolymer of polypropylene and polyethylene [17].

The NIR reflectance spectroscopy would be the proper choice for nondestructive and fast analysis, especially for polymer powder or pellet since samples can be directly measured without further treatments such as hot pressing. It has been investigated for several applications such as the identification of polymers presenting a broad range of morphologies [18] and determination of additive levels in polypropylene pellets [19]. In this publication, NIR reflectance spectroscopy has been examined for following the internal structural change in propylene/ethylene copolymer resulted from the variation of the ethylene content during copolymerization rather than distinct compositional determination in blended polymer or additive level. The change of copolymer structure resulted in clear spectral variations in NIR region, and those spectral features provided selective information to determine the ethylene content. NIR calibration models based on PLS regression were developed for both flake (sample before extrusion) and pellet (sample after extrusion and pelletization) samples. The accuracy of NIR measurement was similar to that of conventional IR method and it has been successfully replaced by the rapid and nondestructive NIR spectroscopic method.

2. Experimental

2.1. Sample preparation and IR spectra

Seventy seven flake and one hundred ten pellet samples were obtained from the actual propylene/ethylene copolymerization process at SK Corporation, Ulsan, Korea, over 16-month period to collect samples from different grades and transition states. The ethylene content ranges for flake and

pellet were 0.0 to 3.6 and 0.7 to 8.7 wt.%, respectively. Pellet samples were spherical in shape with fairly regular size of 3 to 4 mm in diameter, while flake samples were similar in shape and smaller in size (around 1 mm in diameter). To determine ethylene content of samples, IR spectra (4800 to 400 cm^{-1}) were collected using a Nicolet 730 SX FT-IR spectrometer equipped with a KBr beam splitter and DTGS (deuterated triglycine sulfate) detector. A small amount of flake and pellet samples was pressed on a hot plate to produce a thin plastic sheet with thickness from 0.4 to 0.6 mm and corresponding IR spectrum was collected. Each IR spectrum corresponded to an accumulation of 32 scans with a resolution of 2 cm^{-1}. All IR spectra were collected at room temperature. The ethylene content was calculated using the equation

$$\text{wt.\% of ethylene} = A/(A_s \times F),$$

where A is a peak area from 775 to 680 cm^{-1} for impact copolymer and 760 to 700 cm^{-1} for random copolymer, A_s is a peak area from 4482 to 3950 cm^{-1}, and F is a scaling factor for correcting instrumental variations. Each of six standards (thin plastic sheets) of impact and random copolymer with known ethylene content was used to correct instrumental variations daily. The standards were provided from the licenser. The typical IR spectra of two different ranges used for the calculation are presented in Fig. 2. The reproducibility of IR method was ranged from 0.10 to 0.15 wt.% depends on the grade of samples.

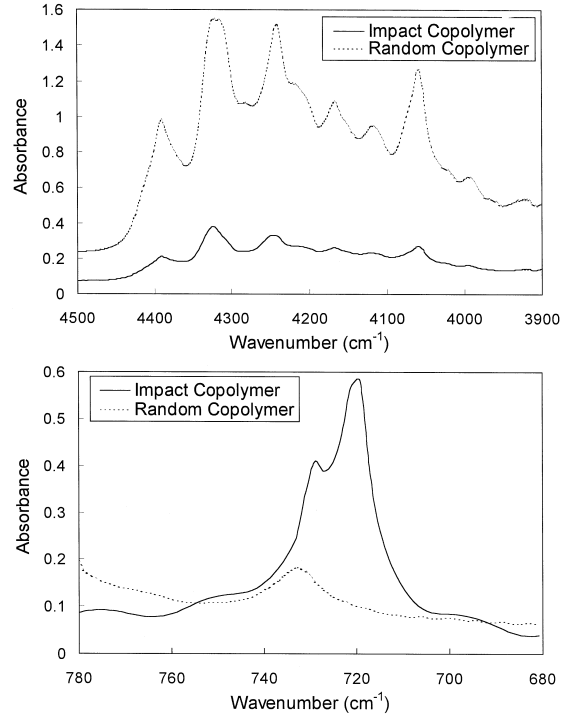

Fig. 2. The typical IR spectra of two different ranges (top: 4500 to 3900 cm^{-1}, bottom: 780 to 680 cm^{-1}) used for the calculation of ethylene content for both impact and random copolymer.

2.2. NIR spectra and data set preparation

NIR spectra were collected over the 9090 to 4000 cm^{-1} (1100 to 2500 nm, resolution: 10 nm with 2 nm data intervals) spectral region using a NIRSystems model 6500 spectrometer (NIRSystems, Silver, MD) equipped with a quartz halogen lamp and four PbS detectors to collect more diffused radiation from samples. The spectrometer was equipped with a sample moving system which allowed scanning of large area of samples approximately 16 cm vertically by continuously moving a sample cell vertically with fixed position of radiation source and detectors. The schematic geometry between a moving sample cell and detectors are shown in Fig. 3. NIR spectra were collected by putting samples into a sample cell (size of window: 3.5 × 26 cm) without any pretreatments. Reference single beam spectra of a ceramic standard were collected for every sample immediately before the collection of the sample single beam spectra and corresponding sample reflectance spectra were obtained. All the spectra were recorded as log $(1/R)$ with respect to a ceramic reference standard. Each sample spectrum was obtained by averaging 32 scans from different areas of samples.

The data sets were prepared for flake and pellet separately. For flake, total 77 spectra were divided into 40 spectra for calibration and 37 spectra for the prediction data set, and total 110 spectra were divided into 56 spectra for calibration and 54 spectra for the prediction data set for pellet. Spectra in both prediction sets were independent from the calibration set and randomly chosen. All data processing including PLS regression and second derivative algorithm

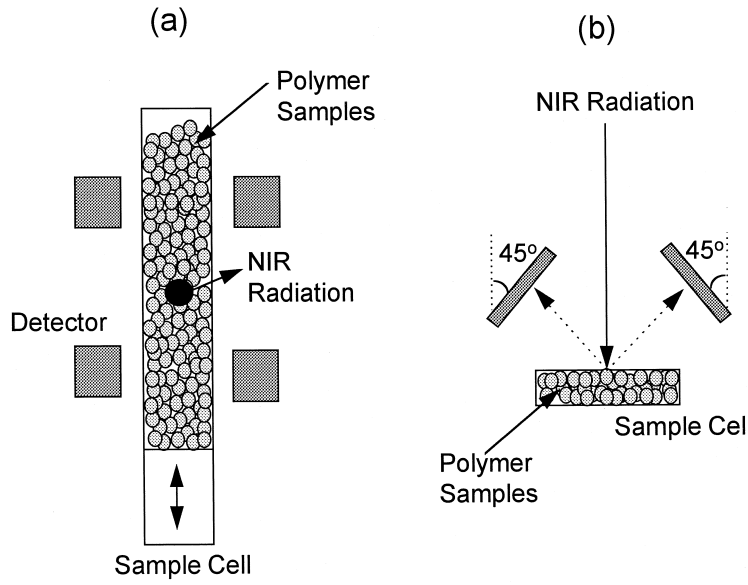

Fig. 3. The schematic geometry ((a): top view, (b): side view) between a moving sample cell and detectors for the collection of reflectance spectra.

were accomplished by using NSAS ver. 3.3 software (NIRSystems).

3. Results and discussion

3.1. NIR spectral features

As depicted in Fig. 1, the more ethylene content results in the more presence of methylene (CH_2) relative to methyl (CH_3) groups in final copolymer structure. Therefore, the NIR spectral variation from the CH_2/CH_3 ratio change in copolymer should be recognized to determine ethylene content. To systematically and simply investigate the spectral variation based on CH_2/CH_3 ratio, normal linear hydrocarbons (*n*-hexane, *n*-heptane, *n*-octane, *n*-dodecane, and *n*-hexadecane) and C6 structural isomers (*n*-hexane, 2-methylpentane, 3-methylpentane, 2,2-dimethylbutane, and 2,3-dimethylbutane) were selected. The corresponding NIR spectra are shown in Fig. 4. The pathlength of measurement was 1 cm. The spectra were linearly offset to enhance qualitative comparison. Overall shapes of all spectra are similar to each other, however reasonable spectral variations are observed from 9090 to 6060 cm^{-1} (1100 to 1650 nm) and 5556 to 4762 cm^{-1} (1800 to 2100 nm) range while the most systematic spectral variations are observed from 8929 to 7874 cm^{-1} (1120 to 1270 nm) range. As increasing the chain length in normal linear hydrocarbons which corresponds to increasing the number of CH_2 groups with fixed number of CH_3 groups, the CH_2 second overtone band at 8264 cm^{-1} (1210 nm) becomes more dominant while CH_3 second overtone band at 8375 cm^{-1} (1194 nm) becomes less dominant. In *n*-hexane spectrum both 8375 and 8264 cm^{-1} bands are observed, however the 8375 cm^{-1} band is clearly decreased in the spectrum of *n*-hexadecane. On the contrary, between C6 isomers, as increasing the number of CH_3 groups (more branched structure) in a given molecular weight, CH_2 second overtone band at 8264 cm^{-1} becomes less dominant while CH_3 second overtone band at 8375 cm^{-1} becomes more dominant. The 8264 cm^{-1} band is clearly decreased in the spectrum of 2,3-dimethylbutane. In C6 isomer spectra, considerable spectral variations are observed from 5556 to 4762 cm^{-1} range, however no systematic variations based on the CH_2/CH_3 ratio are observed.

NIR spectra of actual propylene/ethylene copolymer pellets with zero and 6.7 wt.% ethylene content

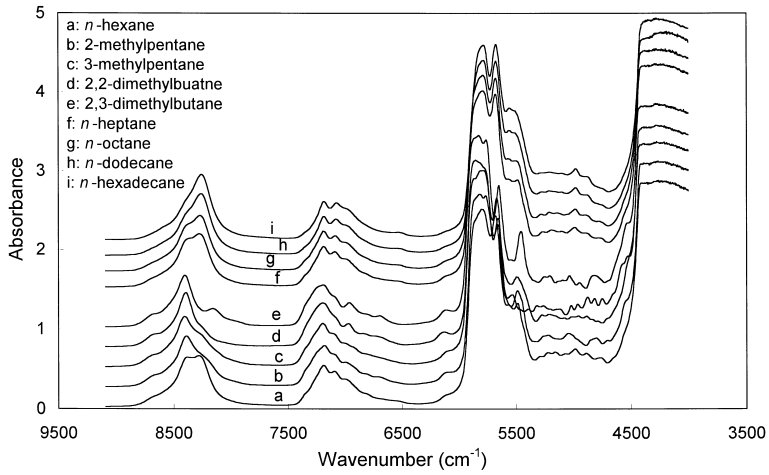

Fig. 4. NIR spectra (9090 to 4000 cm^{-1}) of different chain length of normal linear hydrocarbons and C6 structural isomers.

are shown in Fig. 5. As described above, both 8375 and 8264 cm^{-1} bands are observed like *n*-hexane and clearly relative peak heights are different. For a high ethylene content pellet which corresponds to more CH_2 groups, the 8264 cm^{-1} band is more dominant relative to 8375 cm^{-1} band, while the 8375 cm^{-1} band of CH_3 is more dominant relative to 8264 cm^{-1} band for zero ethylene content pellet. This trend exactly matches the case of normal linear hydrocarbons and C6 isomers. Additionally, minor spectral variations are observed at other NIR region, however no clear and systematic variations are observed.

NIR spectra of all propylene/ethylene copolymer pellet and flake samples used in this study are shown in Fig. 6. Baselines of pellet spectra are more offset from zero reflectance line compared to those of flake due to the loss of whole range of NIR radiation through the space between the packing of pellets. The size of pellet is bigger, so it would be loosely packed compared to flake. Loss of NIR radiation through the loosely packed (void) space by direct

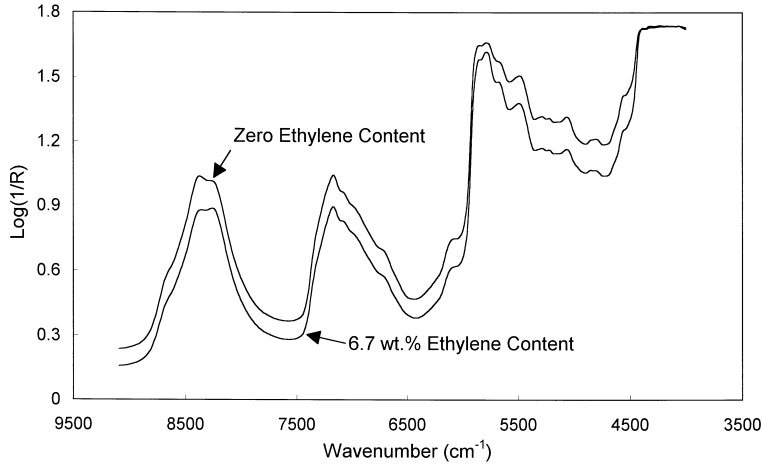

Fig. 5. NIR spectra (9090 to 4000 cm^{-1}) of propylene/ethylene copolymer pellets corresponding to zero and 6.7 wt.% ethylene content.

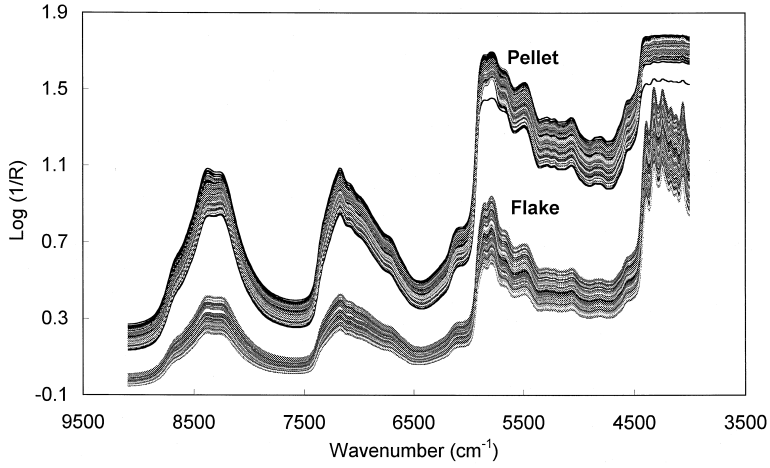

Fig. 6. NIR spectra (9090 to 4000 cm^{-1}) of all propylene/ethylene flake and pellet samples.

penetration of NIR radiation instead of diffused reflectance results in the baseline deviation throughout NIR region. On the contrary, since flake particles are more densely packed compared to pellet, no significant baseline deviations are observed.

The diffuse reflectance spectra of solid propylene/ethylene copolymer are poorly resolved and baseline variations are larger compared to those of liquid NIR spectrum, however, those maintain the same systematic and pertinent quantitative spectral information. The NIR spectral variation found in raw polymer spectra can be successfully utilized to build calibration models. Obviously, instead of simple regression or peak height ratio methods in IR, a multivariate calibration method based on partial least squares (PLS) regression with a proper pretreatment of spectra is essentially required to extract relevant analytical information and quantify the ethylene content.

3.2. PLS calibration models

PLS calibration models were developed with raw and second derivative spectra separately. It is well known that second derivative of raw spectrum helps to enhance the spectral features and remove (or at least reduce) baseline variation [20]. Fig. 7 shows all

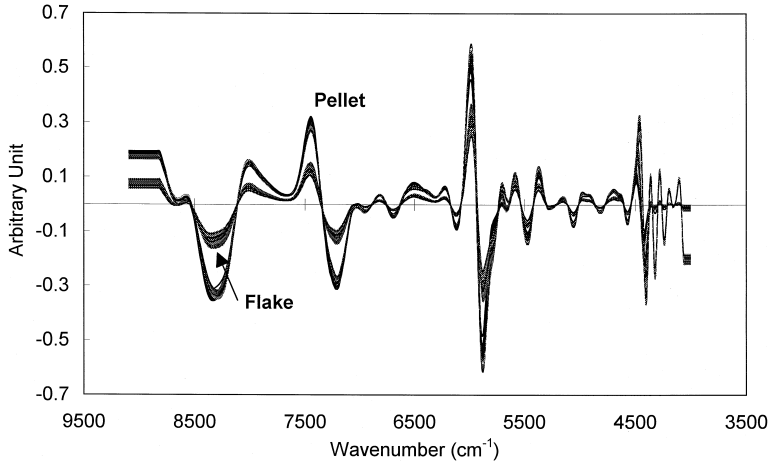

Fig. 7. Second derivative spectra (9090 to 4000 cm^{-1}) of all propylene/ethylene copolymer flake and pellet samples.

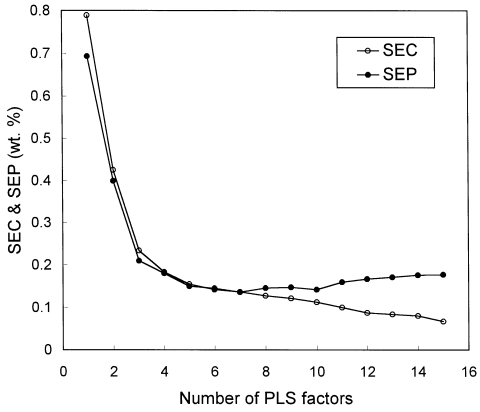

Fig. 8. The effect of number of PLS factors on standard errors of calibration (open circle) and prediction (filled circle) for determination of ethylene content of pellet with second derivative spectra using the 9090 to 6060 cm^{-1} combined with 5556 to 4762 cm^{-1} range.

second derivative spectra of flake and pellet samples corresponding to raw spectra in Fig. 6. The baseline variations are effectively removed and spectral features are enhanced while keeping the same qualitative information.

In the application of the PLS regression, it is generally known that the spectral range and number of PLS factors are critical parameters. It has been previously investigated that the calibration performance varies depends on the spectral range utilized [21]. For flake, two different spectral ranges of 9090 to 6060 cm^{-1} (1100 to 1650 nm) and 9090 to 4762 cm^{-1} (1100 to 2100 nm) were tested. The first spectral range includes 8333 cm^{-1} (1200 nm) and 7143 cm^{-1} (1400 nm) bands, and the second range utilizes the full spectral range except 4762 to 4000 cm^{-1} (2100 to 2500 nm) region. The 4762 to 4000 cm^{-1} region was excluded due to the saturation of NIR radiation which might result in nonlinearity in a calibration model. For pellet, similarly two different spectral ranges of 9090 to 6060, and 9090 to 6060 cm^{-1} combined with 5556 to 4762 cm^{-1} were examined. The absorption bands from 6060 to 5556 cm^{-1} and 4762 to 4000 cm^{-1} were also excluded due to the saturation of NIR radiation.

The optimum number of factors can be identified as the number of factors that gives a minimum standard error of prediction (SEP) for a separate set of validation samples not present in the calibration set. The SEP is an excellent index to represent the ability of a calibration model to predict ethylene contents independently. Fig. 8 shows standard error of calibration (SEC, open circle) and SEP (filled circle) plotted as functions of the number of PLS factors used for the determination of ethylene content of pellet with second derivative spectra using the 9090 to 6060 cm^{-1} combined with 5556 to 4762 cm^{-1} range. As expected, both the SEC and SEP decrease sharply with the initial factors and gradually decrease as more analyte-dependent spectral variation is incorporated into the calibration model. As the number of factors increase further, the SEC continues to drop slightly while the SEP begins to increase at eight PLS factors. This type of trend is typical in many regressions. An increase in the SEP indicates that the data has been overmodelled by incorporating spectral variation into the calibration

Table 1
PLS calibration results of flake and pellet in different spectral ranges using raw and second derivative spectra

Sample	Spectra used	Spectral range (cm^{-1})	Optimal number of PLS factors	SEC (wt.%)	SEP (wt.%)
Flake	Raw	9090–6060	6	0.18	0.17
	Second derivative	9090–6060	7	0.15	0.15
	Raw	9090–4762	5	0.13	0.12
	Second derivative	9090–4762	6	0.12	0.13
Pellet	Raw	9090–6060	5	0.20	0.22
	Second derivative	9090–6060	5	0.16	0.15
	Raw	9090–6060 and 5556–4762	7	0.16	0.22
	Second derivative	9090–6060 and 5556–4762	7	0.14	0.14

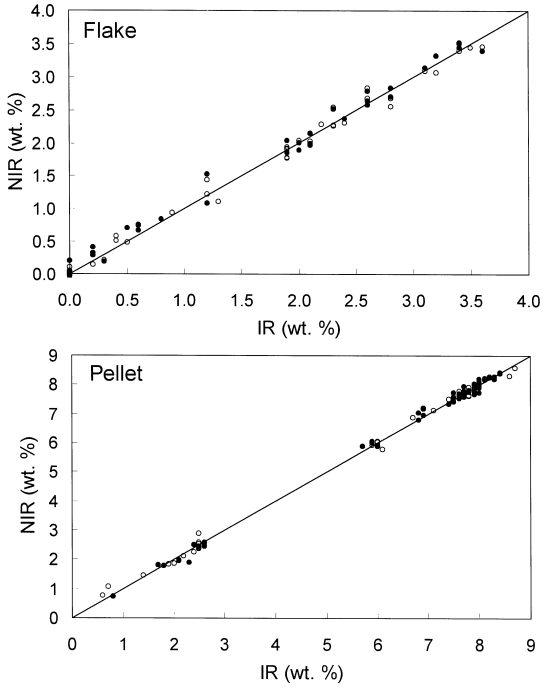

Fig. 9. Scatter plot showing correlation between NIR and IR analyses for flake (top) and pellet (bottom). Open and filled circles correspond to calibration and prediction data, respectively.

model that is not present in the prediction data set. Therefore, in this case, seven PLS factors was chosen as the optimum. The optimal number of PLS factors of other spectral ranges for flake and pellet was determined by generating and examining the same type of plot.

The results of the calibration for flake and pellet employing raw and second derivative spectra in different spectral ranges are summarized in Table 1. In both flake and pellet cases, the wider spectral range with second derivative spectra provided the better calibration models. The spectral variations from the change of ethylene content are present throughout entire NIR region; therefore, the superior calibration model were developed by incorporating more spectral information and eliminating baseline variation by using a second derivative algorithm.

The best calibration models of flake and pellet were obtained by utilizing second derivative spectra of 9090 to 4762 cm^{-1} range with six PLS factors and 9090 to 6060 cm^{-1} combined with 5556 to 4762 cm^{-1} range with seven PLS factors, respectively. Scatter plots showing correlation between NIR and IR analyses for flake and pellet are shown in Fig. 9. Open circles and filled circles represent calibration and prediction data, respectively. In both cases, the calibration and prediction data have good correlation with conventional IR data and many points fall on or close to the unity line. Ethylene contents in the prediction data set were predicted with SEPs of 0.13 wt.% for flake and 0.14 wt.% for pellet. The errors of NIR spectroscopic method for both flake and pellet are comparable and sufficient to replace the reference IR method used to derive the calibration models.

4. Conclusion

NIR spectroscopy has been successfully applied and replaced the conventional IR measurement of ethylene content in propylene/ethylene copolymer. Even though baselines of collected spectra were varying and poorly resolved due to inherent characteristics of diffuse reflectance spectra of solid polymer samples, PLS algorithm effectively extracted the relevant information from NIR spectra and correlated with conventional IR data. With the rapid and nondestructive natures of NIR spectroscopy, the ethylene content can be measured in 2 min without a physical sample pretreatment and the process can be optimally controlled by continuously reflecting the process status from the NIR measurement. Future work will be directed (1) to develop calibration models for other properties of polymer, and (2) evaluate on-line process NIR analyzer system for real-time process control.

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